

Small but Mighty: Code-Switched Javanese and Indonesian Speech Recognition on a Phone

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ABSTRACT

Everyday conversations in East Java mix Javanese and Indonesian inside a single sentence. Commercial recognizers handle Indonesian well but stumble on Javanese words and on the switches. We collect 140 hours of consented, transcribed conversational speech and distill a large multilingual model into a 38 million parameter student that runs in real time on a mid-range phone. The student reaches a 17.8 percent word error rate on code-switched test speech, within 2.1 points of its teacher, and uses a quarter of the memory.

Keywords: speech recognition, code-switching, Javanese, on-device models, distillation

1. Introduction

If you record a lab meeting in Malang, you will hear sentences that start in Indonesian, borrow a Javanese verb, and end with an English acronym. Speech tools rarely expect this, so people who switch languages naturally get worse transcripts.

We focus on the practical question: how small can a model get before code-switched accuracy falls apart, and which training tricks keep it honest [1, 3]?

2. Method

Speakers recorded casual conversations about daily topics with a consent form in both languages. Two annotators transcribed each clip and marked the language of every word. We keep speakers disjoint across the train, development and test splits.

The teacher is a pretrained multilingual encoder decoder fine-tuned on our data. The student is a streaming conformer trained with sequence-level distillation plus a language tag auxiliary loss, which nudges it to notice when the language changes.

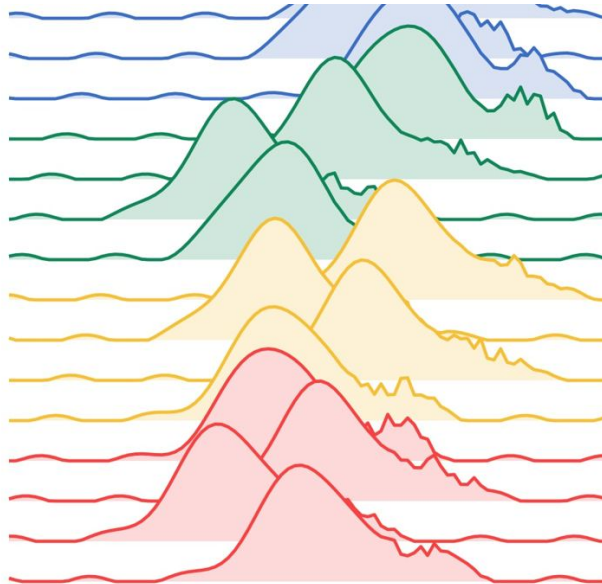


Figure 1. Posterior activity of the language tag head across 16 test utterances. Peaks mark detected switches.

3. Results

Distillation alone closes most of the gap, and the auxiliary loss helps exactly where we hoped: the words right after a switch (Table 1). Latency stays under 200 ms per chunk on the test phone.

System	WER all	WER near switch	Size
Commercial API	31.4%	44.9%	n/a
Teacher	15.7%	22.3%	1.5 B
Student (ours)	17.8%	25.1%	38 M

Table 1. Word error rates on the code-switched test set.

4. Discussion

Errors cluster on Javanese speech levels, where the same meaning uses different words depending on who you talk to. More polite register data would probably help more than a bigger model.

5. Conclusion

A phone-sized recognizer can handle real code-switched speech if the training data sounds like real people. We plan to release the evaluation set with speaker consent.

References

- [1] Gulati, A. et al. Conformer: convolution-augmented transformer for speech recognition. Interspeech, 2020.
- [2] Kim, Y. and Rush, A. M. Sequence-level knowledge distillation. EMNLP, 2016.
- [3] Pratama, R. and Lestari, D. Annotating language switches in conversational speech. Preprint, 2025.
- [4] Radford, A. et al. Robust speech recognition via large-scale weak supervision. ICML, 2023.
- [5] Wijaya, A. Streaming recognition on commodity phones. Zemi talk notes, 2025.